Hemanth Kandimalla

Applied AI Scientist | Agent Alignment & Control

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PROFILE

Applied AI researcher and engineer with 3+ years building domain-adapted models, agent harnesses and training/serving systems. Research focus: alignment and control of agentic models.

EXPERIENCE

Celebal Technologies Sep 2026 - Present
Senior Software Engineer - Data Science | Applied AI Scientist

- Research agent alignment and control through persistent memory, context engineering, workflow/dependency graphs and memory/knowledge graphs.
- Develop domain-specialized SLM/LLM workflows across model adaptation, evaluation, inference efficiency and deployment.

Centre for Development of Advanced Computing (C-DAC) Aug 2023 - Aug 2026
Knowledge Associate | AI Researcher

- Delivered 10 AI proofs of concept; adapted a 3B-parameter LLM and built document ingestion, retrieval/reranking, grounded generation, conversational memory and safety controls behind streaming FastAPI/vLLM services.
- Built agent harnesses with explicit state machines for repository and document workflows: typed tool calls, observations, verifier decisions, bounded repair, test/patch verification, permission gates and rollback.
- Measured serving latency (TTFT/TPOT), throughput and GPU memory; instrumented token/context use and persistent state for long-running workflows.
- Developed adversarial evaluations with red/blue agent loops, prompt-injection and jailbreak tests, groundedness checks and regression gates.

MediaTek Jul 2022 - Jan 2023
Modem Software Engineer Intern

- Validated LTE/5G PHY-MAC behavior using protocol-log analysis and cross-layer hardware/software debugging.

SELECTED ENGINEERING

TrainScale | Built a PyTorch training framework with sequence packing, LoRA/QLoRA and DDP/FSDP paths; checkpointed optimizer, scheduler and RNG state for restart. Implemented Triton/autograd cross-entropy, RMSNorm and RoPE kernels with torch.compile and CUDA Graph execution paths.

SDKAtlas | Built AST-based Python SDK analysis without importing target packages; resolves signatures, inheritance and output-model evidence, reports unresolved cases, and publishes 14 deterministic CSV/JSONL/SQLite artifacts with rollback on failure.

RESEARCH & WRITING

Agentic State | Technical monograph on context, memory and control; separates information, authority and effects to reason about verification and recovery.

The Model Stack | Foundation-model engineering book and research atlas; in development.

TECHNICAL SKILLS

Python • PyTorch • Triton • LoRA/QLoRA • DDP/FSDP • FastAPI • vLLM • MCP/JSON-RPC

EDUCATION

Indian Institute of Technology Jammu | 2021 - 2023
M.Tech, Communications and Signal Processing | CGPA: 8.34/10